A-A-53409
21 March 1988

COMMERCIAL ITEM DESCRIPTION

PLIERS, DENTAL, (Bending)

The General Services Administration has authorized the use of this Commercial Item Description.

Salient characteristics:

This commercial item description covers the following four dental pliers suitable for wire and clasp bending and for shaping metal matrix bands and crown forms.

No. 47
No. 200
No. 114
Jelenko

General requirements. The pliers shall be in accordance with the applicable requirements of Federal Specification GG-I-526 and as specified herein.

Material. Pliers shall be fabricated from either carbon or stainless steel as specified in Table I.

Hardness. After proper stress relieving, the pliers shall have the Rockwell hardness as specified in Table I for the applicable material when determined as specified herein.

Distortion. The pliers shall meet the distortion requirements specified in Federal Specification GG-I-526 when tested as specified herein.

AMSC N/A FSC 6520

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TABLE I. Material, Hardness and Plating.

Material		Rockwell Hardness	Plating
Carbon steel	Carbon content: 0.40 to 1.10 percent Manganese content: 0.45 to 0.90 percent, except when carbon content is over 0.90 percent, the manganese content shall be a maximum of 0.50 percent Phosphorus content: Maximum of 0.045 percent Sulfur content: Maximum of 0.050 percent	C40 to C45, with the exception of the screw or pin. When the carbon content is over 0.90 percent, the hardness shall be C48 to C52.	Plating shall be in accordance with any one of the following three methods: 1. Chromium over nickel plated in accordance with GG-I-526 except the minimum thickness of coating on the box-lock or lap joint area shall be as follows: Copper plus nickel—0.0007 inch. Final nickel————0.0003 inch Chromium————0.000009 inch 2. Nickel plated with minimum thickness of 0.001 inch and maximum thickness of 0.003 inch 3. Nickel plated with a minimum thickness of 0.0004 inch and chromium over-plated with a minimum thickness of 0.00001 inch
Stainless Steel	Class 4, Type 410	C40 to C45, with the exception of the screw or pin.	None
	Class 4, Type 420	C40 to C48, with the exception of the screw or pin.	None

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Design and dimensions. The pliers shall be for use in dental prosthetic procedures suitable for the purpose specified and shall conform to The Defense Personnel Support Center (DPSC) drawings specified herein for the applicable pliers number or name. The pliers shall be furnished in the pliers number or name as specified: Number 47 shall be suitable for dental and optical wire bending and clasp adjustment. Number 200 shall be suitable for wire bending (sharp bends) and clasp adjustment (enlarging or reducing clasps). Number 114 shall be contouring pliers suitable for shaping copper bands, steel matrix bands, and metal temporary crown forms. The Jelenko pliers shall be suitable for holding and bending wire having four grooves (three numbered grooves) to accommodate from 12 to 20 gage wire. The inside of one beak shall be flat and the inside of the other beak shall be rounded. The grooves beginning at the tip, shall be suitable for the following wire sizes (Brown and Sharpe gage):

Unnumbered groove - For 0.040 inch to 0.032 inch diameter wire (18 to 20 gage).

No. 1 groove - For 0.050 inch to 0.040 inch diameter wire (16 to 18 gage).

No. 2 groove - For 0.064 inch to 0.057 inch diameter wire (14 to 15 gage).

No. 3 groove - For 0.080 inch to 0.072 inch diameter wire (12 to 13 gage).

Style and Drawing Number.

Pliers Designation	DPSC Drawing No.
No. 47	1832
No. 200	20659
No. 114	24314
Jelenko	24294

Locks. The joint of the pliers shall be box-lock or lap-joint type, conforming to the requirements of Federal Specification GG-I-526 except for No. 200 which shall be box-lock type only. The joint shall not bind, and shall not exhibit play or looseness in any position of the joint from a full open to a full closed position. The edges of the joints shall be properly beveled to prevent pinching of or injury to tissue. The joint of the pliers shall be properly centered and uniform in cross-section when tested as specified herein.

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Finish. The beaks and head of the pliers shall be smoothly polished. In addition to the requirements of Federal Specification GG-I-526, the pliers shall have a mirror, satin, matte, sandblast, or a smooth, nonreflecting surface finish. The matte finish shall exhibit a uniform nondirectional surface texture equivalent to that produced by lightly blasting the prepared surface with a fine abrasive under pressure. The outside surfaces of the joint shall present a smooth surface when the pliers are in the closed position. The outside surface of the pin or screw shall be flush with the outside contour of the pliers in the open or closed position.

Plating. Plating shall be in accordance with Table I, as applicable for the material specified.

Corrosion resistance. All pliers shall meet the boil test requirements of Federal Specification GG-I-526 when tested as specified herein. Pliers fabricated from stainless steel shall also meet the cooper sulfate test requirements of Federal Specification GG-I-526 when tested as specified herein.

Identification marking. Marking shall be in accordance with applicable requirements of Federal Specification GG-I-526. Instruments of foreign manufacture shall also be marked with the word "Stainless". All marking shall be legible and permanent.

Workmanship. The plating of the pliers shall be complete, non-porous, continuous and free from flaking or blisters. The jaws of the pliers shall be properly aligned. The joint shall be free from stiffness and excessive play and shall have properly beveled edges. The pliers shall be free from cracks or other defects which may detract from their appearance or impair their serviceability, proper functioning, or intended use.

Unit. Each (EA). One pliers of the number or name specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.5 (percent defective) and an inspection level of II. Unit of product for examination purposes shall be one pliers.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as specified herein. Acceptance number shall be zero for all sample sizes. Lots for hardness testing shall be as specified herein. The unit of product for testing purposes shall be one pliers. Where feasible, the same unit of product shall be used for the determination of two or more test characteristics.

Sampling Criteria for Tests.

Characteristic	Inspection level
Chemical analysis	(See sampling of raw material)
Hardness	S-2
Distortion	S-2
Lock stiffness	S-2
Corrosion resistance	S-2

Sampling of raw material. A sufficient quantity from each lot of metal used in the construction of the pliers shall be used for composition analysis.

Examination. The pliers shall be examined to determine compliance with requirements contained in this specification. Examination shall include, but shall not be limited to, the defects listed below.

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Defects

Style and design not as specified.
Finish not as specified.
Plating of plated pliers not free from flaking or blisters.
Plating of plated pliers incomplete, porous, or not continuous.
Jaws do not align properly.
Joint of pliers not free from stiffness.
Joint of pliers has excessive play.
Edges of the joint of pliers sharp (not properly beveled).
Pliers not free from cracks.
Identification marking not present, legible, or complete.
Pliers not permanently marked with the name or trademark of the contractor.

Tests. Tests shall be conducted to determine compliance with the requirements of this specification.

Chemical analysis. In the absence of certificates of quality for the raw material, chemical analysis shall be conducted in accordance with method 111 or 112 of FED-STD-151. In case of dispute, method 111 shall govern.

Hardness. Hardness shall be determined in accordance with American Society for Testing and Materials (ASTM) Designation E18 (Rockwell) or E92 (Vickers or Diamond Pyramid).

Distortion. The pliers shall be subjected to the distortion test as specified in Federal Specification GG-I-526, except that the thickness of the wood or strap shall be 3/32 to 1/8 inch.

Lock stiffness. The pliers shall be subjected to the lock stiffness test as specified in Federal Specification GG-I-526.

Corrosion resistance. The pliers shall be subjected to the copper sulfate and the boil tests as specified in Federal Specification GG-I-526.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

National Stock Number	Item Identification
6520-00-079-9574	PLIERS, DENTAL, No. 114
6520-00-543-5350	PLIERS, DENTAL, NO. 47
6520-00-543-4150	PLIERS, DENTAL, NO. 200
6520-00-543-4550	PLIERS, DENTAL, Jelenko

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Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6520-2048

Location: NCR Disk #4

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53409	PLIERS, DENTAL, (Bending)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording.

b. Recommended Wording.

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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